SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – Short Answers

Course Code: ITA03	**Course Title:** Mobile computing
CO Assessed: CO1	**Assessment:** Assessment Tool 2 – Short Answers
Weightage: 30 % of CIA	**Total Marks:** 100 Marks

CO1 Statement: Analyze the mobile network components, mobile base station, mobile infrastructures and protocols in cellular systems.

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Section / Batch: 2023	**Date:** 20/10/2026

Code of Conduct

I B. Lavanya / 192372046 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: B. Lavanya

Instructions

- This test contains 50 questions, Total: 100 Marks.
- Duration: 1 hour.

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

Questions and Answers

Q1. A GSM operator sets carrier spacing at 200 kHz and rolls out 100 carriers, each carrying 8 time slots for voice traffic. Determine the total number of simultaneous voice channels this deployment can offer.

Given: number of carriers = 100; time slots per carrier = 8.

Formula: Total channels = number of carriers x slots per carrier.

Solution: Total channels = 100 x 8 = 800.

(Check: spectrum used = 100 x 200 kHz = 20 MHz.)

Answer: 800 simultaneous voice channels.

Q2. A drive test captures a serving signal at -60 dBm alongside co-channel interference at -92 dBm. The RF team needs the Carrier-to-Interference ratio (C/I) in dB to assess link quality.

Given: C = -60 dBm; I = -92 dBm.

Formula: C/I (dB) = C (dBm) - I (dBm), since subtracting dBm values gives a ratio in dB.

Solution: C/I = -60 - (-92) = -60 + 92 = 32 dB.

Answer: C/I = 32 dB (a good link, as the signal is about 1585 times stronger than the interference).

Q3. A drive test records a vehicle traveling at 100 km/h while camped on a 900 MHz carrier. Estimate the Doppler shift the receiver experiences.

Given: v = 100 km/h; f = 900 MHz; speed of light c = 3×10^8 m/s.

Formula: $f_d = v / \text{lambda} = (v \times f) / c$.

Solution: v = 100 x 1000 / 3600 = 27.78 m/s.

lambda = c / f = $(3 \times 10^8) / (900 \times 10^6)$ = 0.333 m.

f_d = 27.78 / 0.333 = 83.3 Hz.

Answer: Maximum Doppler shift is about 83.3 Hz.

Q4. A handset transmits at 8 W, but the signal experiences 33 dB of path loss on its way to the base station. Estimate the power the base station receiver actually sees, in dBm.

Given: P_t = 8 W; path loss = 33 dB.

Formula: P (dBm) = 10 $\log_{10}$(P in mW); P_r (dBm) = P_t (dBm) - path loss (dB).

Solution: P_t = 8 W = 8000 mW, so P_t (dBm) = 10 $\log_{10}$(8000) = 39.03 dBm.

P_r = 39.03 - 33 = 6.03 dBm.

(Check: $10^{0.603}$ = 4.0 mW, which equals 8 W / 1995.)

Answer: Received power is about 6.03 dBm (approx. 4 mW).

Q5. A market study notes that 1200 subscribers each place 4 calls per hour, averaging 2.5 minutes per call. What total offered traffic, in Erlangs, does this represent?

Given: subscribers = 1200; call rate = 4 calls/hour each; average holding time = 2.5 min.

Formula: Traffic A (Erlangs) = call arrival rate x average holding time.

Solution: Total calls per hour = 1200 x 4 = 4800.

Holding time = 2.5 / 60 hour = 0.04167 hour.

A = 4800 x 0.04167 = 200 Erlangs.

Answer: Offered traffic = 200 Erlangs.

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand